

UNIVERSITY OF BARISHAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATABASE PROJECT REPORT

Bangladesh Bank Prize Bond Management System

A Report Submitted in Partial Fulfillment of the Requirements for the Course:

CSE-2102: Database Management System Lab

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1. Overview of the Project

Prize Bond is a one kind of lottery system provided by Bangladesh Bank. This draw will organize after every 3 months later. **The Prize Bond Management System** is a software solution designed to manage and automate the process of prize bond purchases, store for automate checking, prize draws, and notifications.

The system allows users to register, purchase bonds, check results, and receive notifications when they win.

It ensures accuracy, transparency, and security while simplifying the prize bond management process. It reduces users time for checking bond number from newspaper.

Existing System:

1. User should maintain manual records of their bonds which can be lost, damaged or forgotten.
2. Draw result published in newspaper or govt. website and users should check each of bond numbers manually.
3. Users must stay alert when it published on newspaper.

Proposed System:

1. Users maintain a digital record of their prize bonds.
2. The system automatically checks users bond number.
3. Users receive notification directly if they have won a prize.
4. Users can also purchase prize bond & give us feedback.



2.Objectives of the Project

The main objective of this project is to design the database for the Bangladesh Bank Prize Bond. And secondly to develop a website by how a user can efficiently check, purchase their prize bond from here.

Key goals include:

1. Users Authentication log in by hash password.
2. Allow users to add their prize bond.
3. Enable real-time checking of draw results.
4. Provide notifications to users on winnings.
5. Users can purchase new prize bond.
6. Take feedback from users.

3.Motivation for the Project

- ❖ Traditional methods rely on newspapers and government websites.
- ❖ Vast Marketplace around 10 lacs users holding prize bond.
- ❖ Users must manually check each bond, increasing the risk of missing wins.

4.Project Requirements:

The system must manage users, bonds, purchases, notifications, and prize draws efficiently. Now describe all functional requirement shortly:

User Management:

Users must be able to register an account with personal details (Full name, Email, Phone, Address etc.)which add to a profile table & users must be able to log in and logout securely.

Prize Bond Management

Every user has zero to many prize bonds. Every prize bond has unique number. Some prize bond has owner and some prize bonds are available for purchase. They are store in separate table. User able to purchase and show their bond.

Prize Draw Management

After every draw organized from Bangladesh Bank, draw result must be updated here then system automatically check which bond are win and who are the owner.

Notification & Feedback

When system automatically check and found winner then notify him. The system should send notifications via email or SMS. Users can give feedback about any issue.

Transaction & Payment

Users must be able to pay for prize bonds using multiple payment methods. The system generates receipts for purchase.

5.Scope of the Project

- User registration and profile management.
- Prize bond purchasing and ownership tracking.
- Automated notification system for winning bonds.
- Prize draw result management.
- Collecting Feedback.

6.Designing Entity-Relationship (ER) Diagram

Requirements:

The Prize Bond Management System allows **users** to securely register and maintain their information, including personal details such as name, phone number, address, and bank account information for prize deposits. The system enables users to purchase prize bonds from the **bonds** listed, and add bonds are linked to their profiles. After publishing a **draw** result, the system regularly conducts prize draws, and users can check the **draw results** to see if their bonds have won. If a user wins, they receive **notifications** about their winnings. The system categorizes prizes based on predefined **prize categories**. Additionally, the platform features a **feedback** module that allows users to share their experiences, helping to enhance the system's overall user experience.

Step 1: Identify the Entities Required:

- ☐ **users**
- ☐ **bonds**
- ☐ **draws**
- ☐ **draw_results**
- ☐ **notification**
- ☐ **prize_category**
- ☐ **feedback**

Step 2: Identify the Attributes & Primary Keys

1. users(user_id (PK), name, phone ,email,password, address, bank_account)
2. bonds(bond_id (PK), bond_num, status, added_date,)
3. draws(draw_id (PK), draw_date, draw_round)
4. draw_results(result_id (PK), draw_id, bond_num)

5. prize_category(cat_id (PK), category, prize_amount)
6. notification(notify_id (PK), message, status, created_at)
7. feedback(feedback_id (PK), message)

Step 3: Identify Relationships

1. users---purchase---bonds
2. users---add---bonds
3. draws---have---draw_results
4. draw_results---categorize---prize_category
5. draw_results---check----bonds
6. users---receive---notification
7. users---give---feedback

Step 4: Identify the Cardinality Ratio and Participation

1. users---purchase---bonds
 - Relationship: A user can purchase multiple bonds.
 - Cardinality: 1:N



2. users---add---bonds

- Relationship: One user can add multiple bonds.
- Cardinality: 1:N



3. draws ---have---draw_results

- Relationship: One draws have many winner bonds.
- Cardinality: 1:N



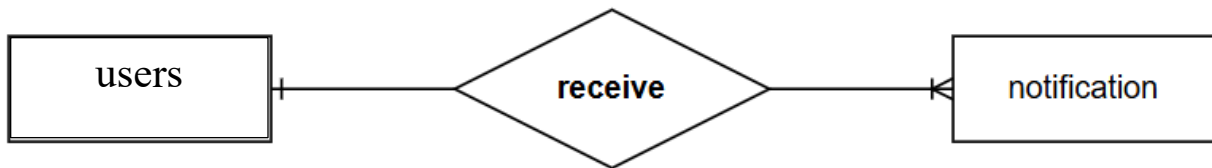
4. draw_results-----check_bond-----bonds

- Relationship: Many winner bond number are checked against many bond number of users.
- Cardinality: 1:N



5. users---receive---notifications

- Relationship: A profile can get more than one notification
- Cardinality: 1:N



6. draw_results---categorize---prize_category

- Relationship: One draw results have many category prize.
- Cardinality: 1:N



7. profiles---give---feedback

- Relationship: A profile can submit multiple feedback messages.
- Cardinality: 1:N



Step 5: Draw Entity Relationship Diagram

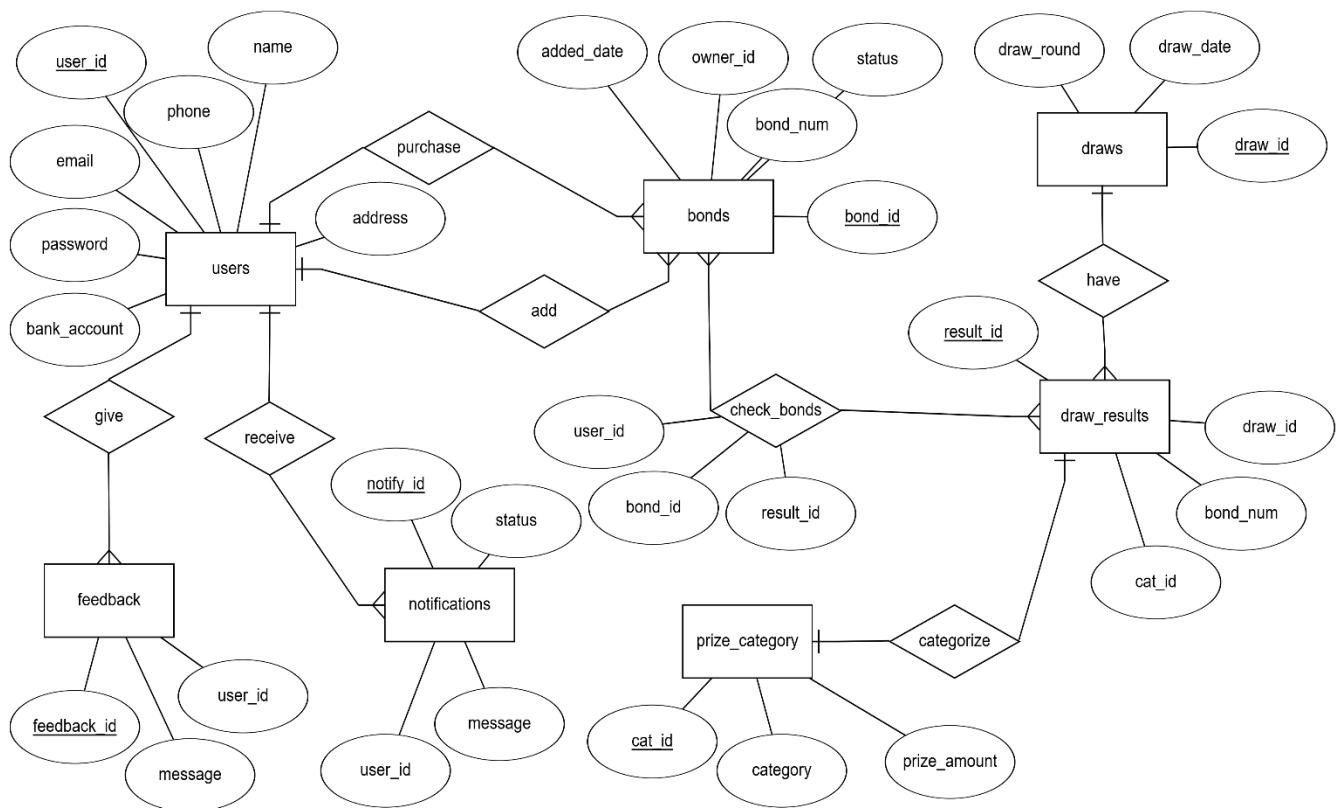


Fig-ERD diagram

6.Reduction to database schema:

1. users(**user_id** (PK), name, phone ,email,password, address, bank_account)
2. bonds(**bond_id** (PK), bond_num, status, added_date,owner_id)
3. draws(**draw_id** (PK), draw_date, draw_round)
4. draw_results(**result_id** (PK), draw_id, bond_num,cat_id)
5. prize_category(**cat_id** (PK), category, prize_amount)
6. notification(**notify_id** (PK), message, status, created_at, user_id)
7. feedback(**feedback_id** (PK), message,user_id(FK))
8. check_bonds(**check_id**(PK),user_id,result_id,bond_id)

7. Schema Diagram:

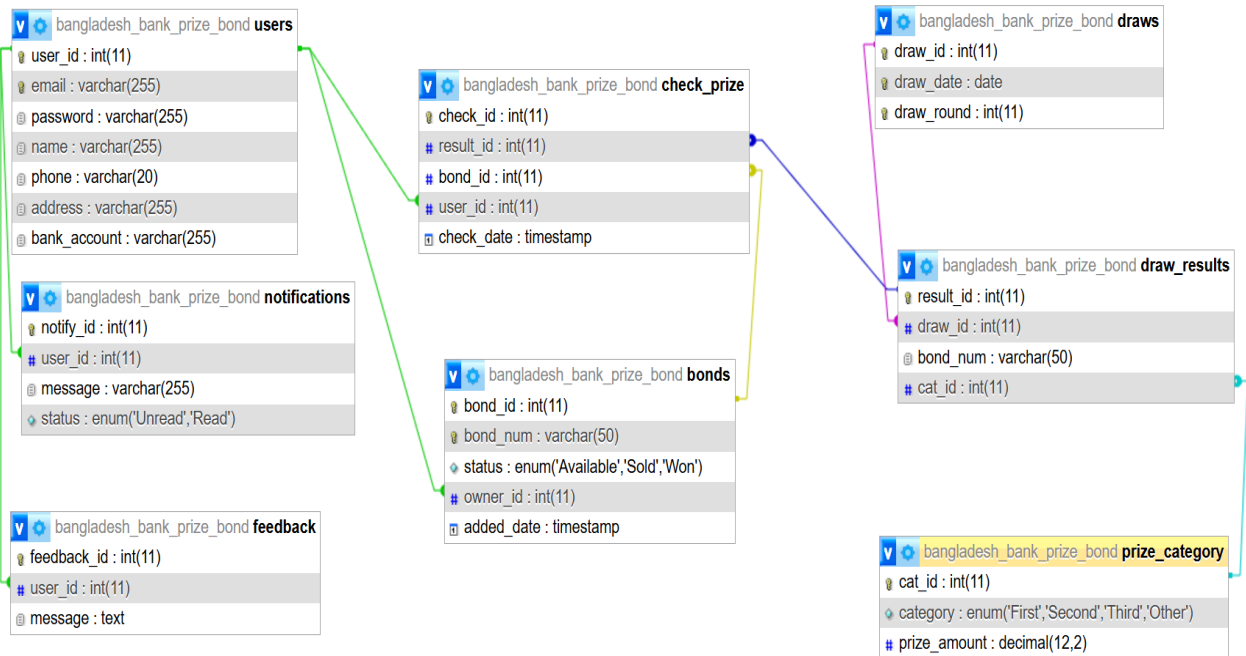


Fig-Schema diagram

Future Work

- ✦ Develop a Full-stack website to implement this database.
- ✦ Developing a mobile application for easy prize bond checking.
- ✦ Enhancing security measures using blockchain technology.
- ✦ Supporting multi-currency and international bonds.

Conclusion

The Prize Bond Management System offers an efficient and automated way to manage prize bond. User just put their bond number here and this system will automatically check bond number. If user win it notify him. Future improvements can include additional features like online prize claims and mobile application support to enhance user experience.